

The background of the slide is a blurred image of a computer monitor. The screen shows what appears to be a code editor with various lines of code in different colors (blue, green, white) on a dark background. At the bottom of the screen, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, serving as a backdrop for the text.

C#

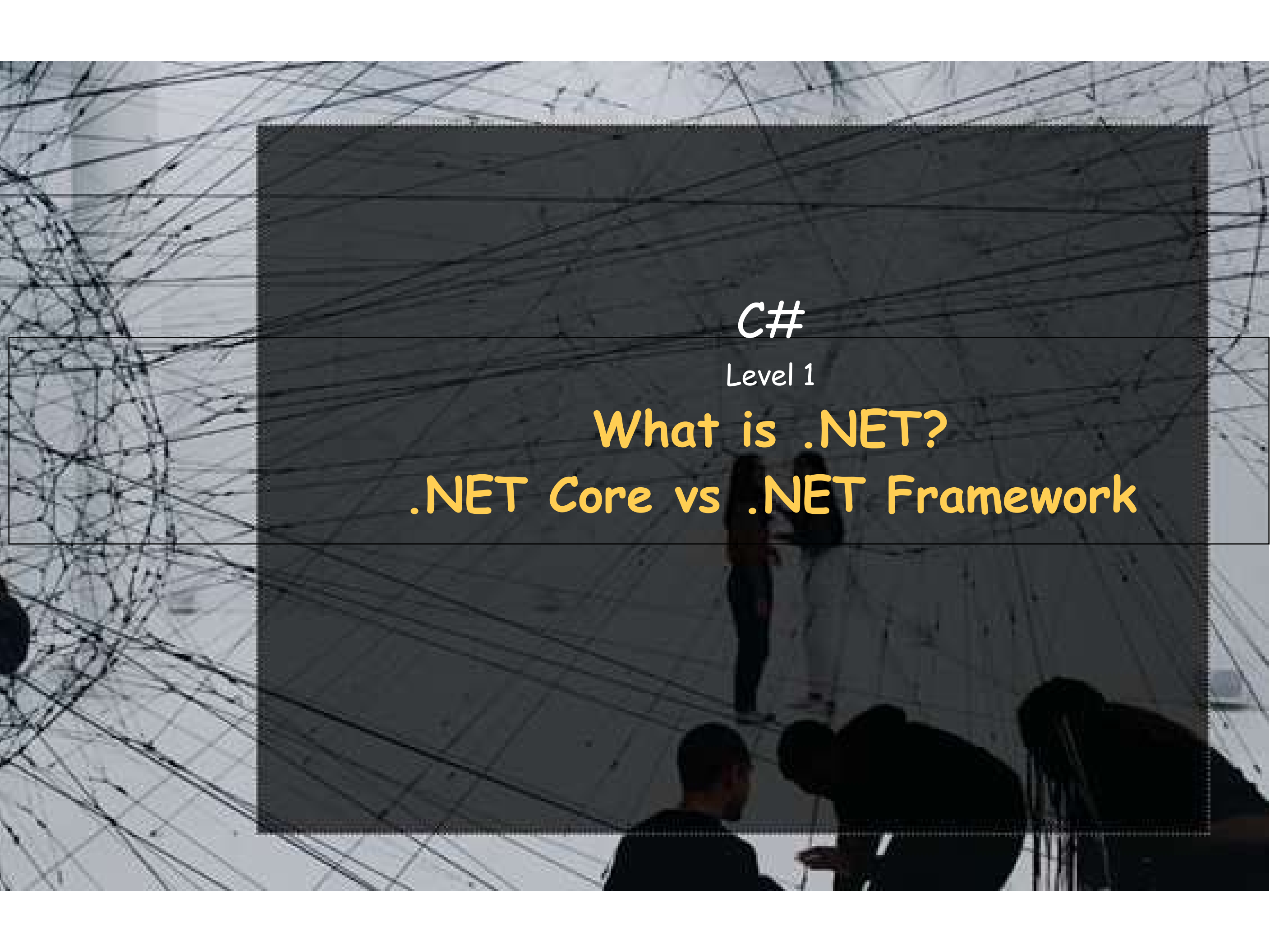
Level 1

What is .NET?

.NET Core

vs

.NET Framework



C#

Level 1

What is .NET?

.NET Core vs .NET Framework

Definition

.NET

-  .NET is a free, open-source, cross-platform developer platform maintained by Microsoft and the .NET community.

-  It provides languages, runtime environments, libraries, development tools, and frameworks for different application types.

-  Common .NET languages include C#, F#, and Visual Basic.

-  .NET is a platform, not a programming language.

.NET Framework

-  .NET Framework is an older implementation of the .NET platform designed primarily for Windows applications.

-  It provides a runtime, class libraries, and technologies for Windows desktop and web applications.

.NET Core

-  .NET Core was Microsoft's cross-platform, open-source, and modular implementation of .NET.

-  It supported Windows, Linux, and macOS.

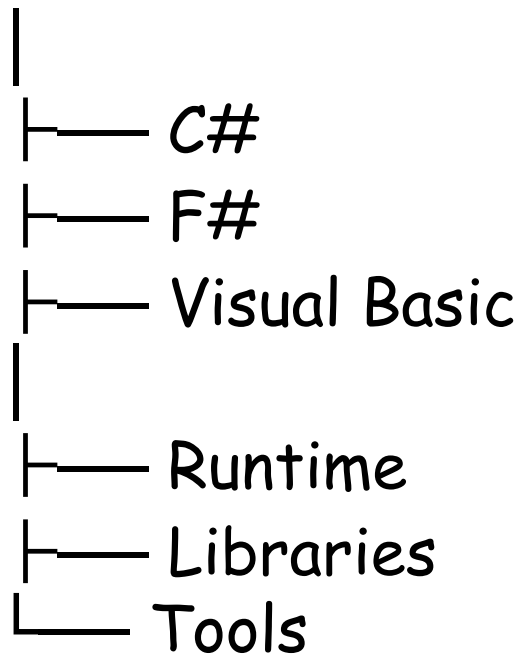
-  The name ".NET Core" was used through .NET Core 3.1.

-  Starting with .NET 5, Microsoft unified the platform under the name .NET.

Explanation

.NET Platform

.NET Platform



➡ Developers can use different .NET languages while working with the same overall platform and libraries.

.NET Evolution

.NET Framework

+

.NET Core



Modern .NET
(.NET 5+)

Explanation

- ▣ .NET Framework → older, Windows-focused implementation

- ▣ .NET Core → cross-platform implementation used before the unified .NET line

- ▣ Modern .NET → unified platform beginning with .NET 5

- ▣ .NET Standard

- ▣ .NET Standard was a formal specification of APIs that different .NET implementations could support.

- ▣ It helped libraries target common APIs across different .NET implementations.

- ▣ For new modern .NET development, it is generally less central because modern .NET versions share a common API surface.

- ▣ .NET Architecture

- ▣ Runtime executes applications and provides services such as memory management, garbage collection, and exception handling.

- ▣ The Base Class Library provides reusable APIs.

- ▣ .NET supports multiple programming languages.

Procedures

1. Choose the application type.
2. Choose a .NET language such as *C#*.
3. Choose the appropriate .NET framework or technology.
4. Use the .NET runtime and libraries.
5. Build and run the application on the target platform.

💡 Example

Suppose you want to create a web API:

C#



ASP.NET Core



.NET



Windows / Linux / macOS

➡ Result:

You can build a cross-platform web API using *C#* and ASP.NET Core.

Note

◆ C# ≠ .NET

▨ C# → Programming language

▨ .NET → Development platform

▨ ASP.NET Core → Web framework built on .NET

C#



Programming Language

.NET



Platform

ASP.NET Core



Web Framework

◆ Modern .NET versions include .NET 5, .NET 6, .NET 7, .NET 8, .NET 9, and .NET 10.

⊘ Warning

- ◆ Do not say ".NET is a programming language."
 - ➡ .NET is a platform.
- ◆ Do not treat .NET Core and modern .NET as separate current platforms.
 - ➡ .NET Core was the earlier name for the cross-platform line before the unified .NET platform.
- ◆ Do not assume .NET Framework is the same thing as modern .NET.
 - ➡ They are different implementations/generations.



Comparison

▨ .NET Framework

- ✓ Older implementation
- ✓ Primarily Windows
- ✓ Large existing ecosystem
- ✓ Used heavily by legacy applications
- ✓ Supports technologies such as WPF and Windows Forms

▤ .NET Core

- ✓ Cross-platform
- ✓ Open-source
- ✓ Modular
- ✓ High performance
- ✓ Older name for the modern cross-platform .NET line

▤ Modern .NET

- ✓ Unified platform
- ✓ Cross-platform
- ✓ Open-source
- ✓ Actively developed
- ✓ Used for most new .NET development

✦ Summary

- ✓ .NET is a development platform created and maintained by Microsoft and the .NET community
- ✓ It supports languages such as C#, F#, and Visual Basic
- ✓ .NET Framework is the older Windows-focused implementation
- ✓ .NET Core was the earlier cross-platform implementation
- ✓ Starting with .NET 5, Microsoft unified the platform under the name .NET
- ✓ Modern .NET runs on Windows, Linux, and macOS
- ✓ .NET can be used for desktop, web, cloud, mobile, gaming, IoT, and AI
- ✓ C# = Language
- ✓ .NET = Platform
- ✓ ASP.NET Core = Web Framework

Thank you 😊